Quantification: 682_z29
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	15
interior lacuna count	13
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 13)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1650.75	1511.5	948.377	px^2
major_axis_length_px	lacuna shape	74.0009	74.236	22.3194	px
minor_axis_length_px	lacuna shape	27.9394	29.9101	9.5919	px
aspect_ratio	lacuna shape	2.805	2.6218	0.775	unitless
eccentricity	lacuna shape	0.9144	0.9241	0.0615	unitless
solidity	lacuna shape	0.8593	0.8693	0.0551	unitless
roots_count	headline	7.8462	8	3.9549	count
ring_length_r30_px	headline	312.923	313	123.625	px
ring_length_r60_px	reported	777.462	842	298.445	px
owned_length_px	ownership-dependent	2261.21	1997.69	1901.22	px
edge_count	ownership-dependent	99.2308	73	87.8798	count
mean_edge_length_px	ownership-dependent	23.6307	22.8286	3.5199	px
perimeter_px	normalised	186.054	202.693	70.3766	px
ring_area_r30_px2	normalised	7571.77	7711	2170.77	px^2
ring_area_r60_px2	normalised	19073	19480	4883.71	px^2
in_frame_fraction_r30	normalised	0.9952	1	0.0173	unitless
in_frame_fraction_r60	normalised	0.9839	1	0.0537	unitless
ring_density_r30	normalised	0.0413145	0.0409894	0.0106571	px^-1
ring_density_r60	normalised	0.040457	0.0406043	0.00946471	px^-1
roots_per_100px_perimeter	normalised	4.2332	4.2023	1.3649	per 100 px
ring_attached_length_r30_px	network v2	279.923	278	125.338	px
ring_attached_length_r60_px	network v2	639.615	639	314.74	px
ring_length_w_r30_px	network v2	350.515	338.409	136.278	px
ring_length_w_r60_px	network v2	874.235	926.056	334.506	px
sholl_crossings_r10	network v2	7.5385	7	2.9613	count
sholl_crossings_r20	network v2	9.6923	11	3.924	count
sholl_crossings_r30	network v2	11.2308	10	3.8113	count
loop_count	loops, report only	10.2308	5	13.6757	count
loop_edge_share	loops, report only	0.2228	0.161	0.2153	share
ring_width_mean_r30_px	width	7.1965	7.1922	0.5417	px

Per field
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measure	value	units
analysed_area_px2	1.02273e+06	px^2
total_skeleton_length_px	37663	px
canalicular_length_density_per_px	0.036826	px^-1
junction_count	824	count
junction_density_per_px2	0.00080569	px^-2
lacuna_count	15	count

lacunae_per_px2	1.467e-05	px^-2
skeleton_component_count	236	count
mean_component_length_px	159.589	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0415821	px^-1
field_density_without_flagged_per_px	0.0371185	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	513	px
unattached_component_count	32	count
canalicular_length_density_attached_per_px	0.036826	px^-1
field_length_density_with_unattached_per_px	0.0373276	px^-1
loop_count	165	count
loop_edge_share	0.3406	share
edge_count	1826	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6905	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	35153	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2083	86.508	34.896	2.479	0.915	0.8266	1.5435	13.4983	99.4023	True
2	2213	89.1614	32.1486	2.7734	0.9327	0.8778	-1.1291	134.914	921.292	False
3	446	49.6136	12.2658	4.0449	0.969	0.7866	-1.4445	249.987	449.359	False
4	3140	120.617	34.4194	3.5043	0.9584	0.8885	1.5609	431.048	220.093	False
5	2837	88.9183	41.496	2.1428	0.8844	0.9229	-1.5453	532.043	493.055	False
6	1634	75.9278	30.0551	2.5263	0.9183	0.8211	-1.4797	555.483	357.187	False
7	3781	121.501	43.6088	2.7862	0.9334	0.8771	1.4272	590.482	55.0987	True
8	1399	81.52	23.1918	3.515	0.9587	0.8541	1.5542	716.135	181.57	False
9	663	61.7115	15.8655	3.8897	0.9664	0.7408	-1.1603	728.596	861.06	False
10	513	41.8409	17.3844	2.4068	0.9096	0.8607	1.3405	830.565	75.5867	False
11	1130	64.3164	23.6688	2.7173	0.9298	0.8835	1.4062	842.292	125.237	False
12	1244	49.7173	33.3792	1.4895	0.7411	0.9325	-1.542	904.6	760.807	False
13	2966	92.1224	41.6334	2.2127	0.8921	0.895	-1.4899	920.401	307.385	False
14	1624	72.5441	29.7651	2.4372	0.9119	0.8476	-0.945	978.497	437.615	False
15								548.08	212.663	False

Network per lacuna

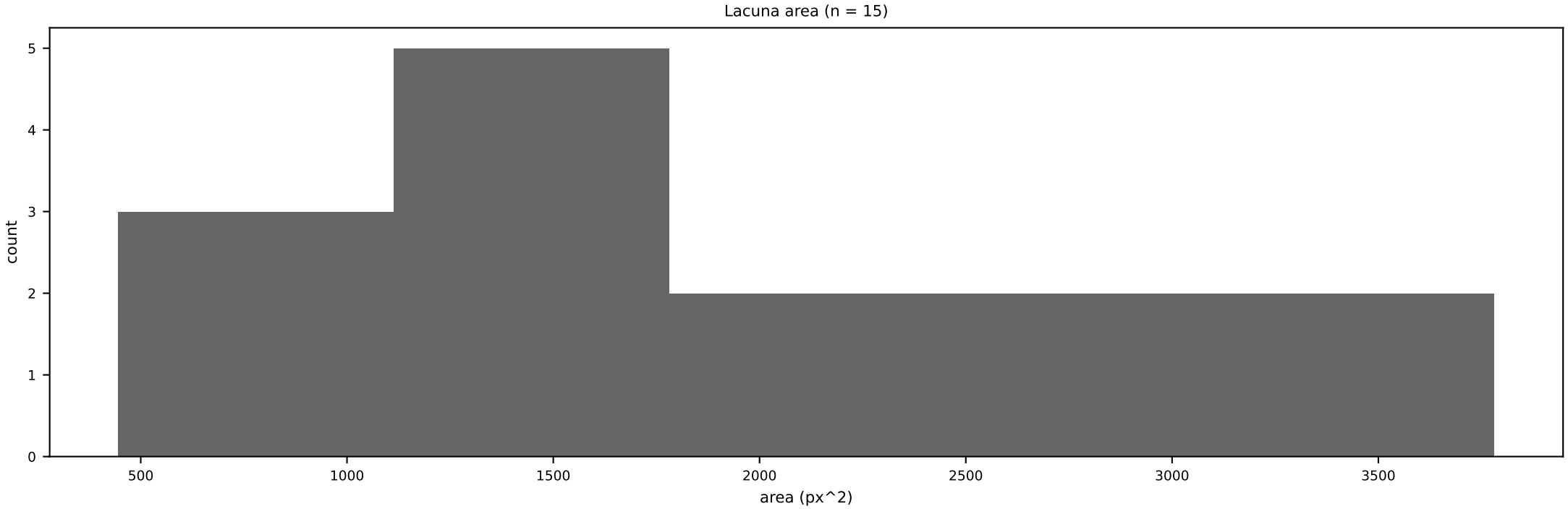
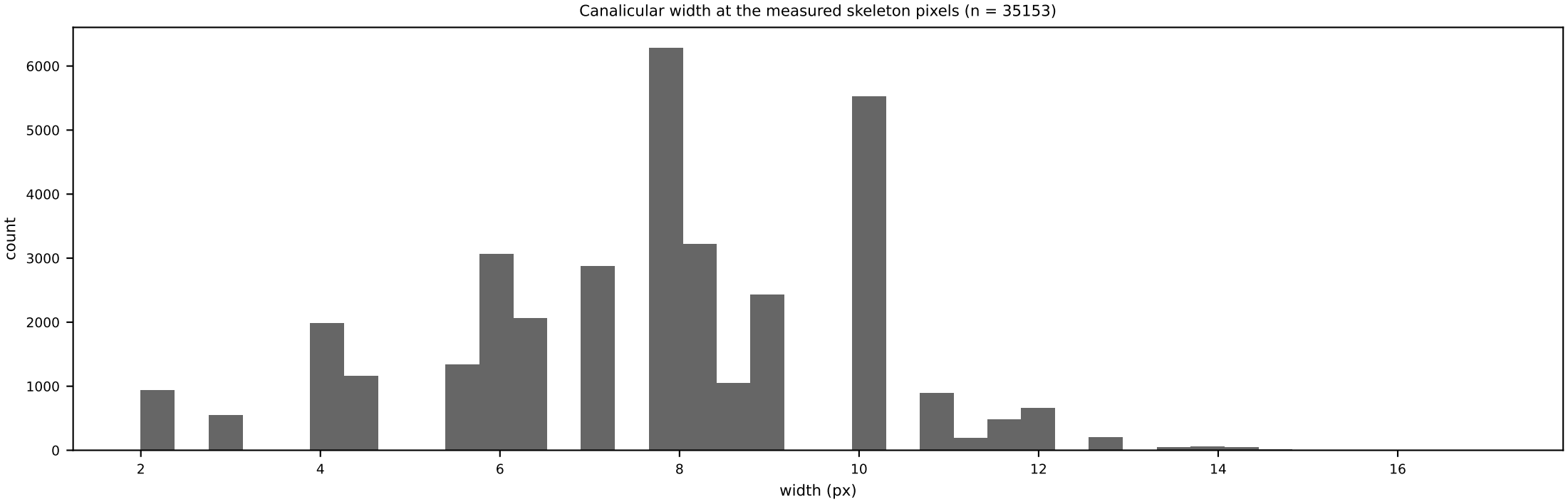
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	192	359	1495.72	79	18.9332	
2	17	536	1315	6300.68	276	22.8286	7.2892
3	6	361	940	4633.43	213	21.7532	7.3901
4	12	476	1282	4171.5	217	19.2235	7.7077

5	7	313	842	2980.13	118	25.2553	7.1922
6	9	432	897	2686.36	119	22.5744	7.3207
7	11	392	821	1426.67	64	22.2918	7.4798
8	8	376	895	1997.69	73	27.3657	6.2524
9	9	214	646	2834.25	116	24.4332	6.9231
10	5	174	466	509.161	22	23.1437	6.8561
11	5	190	476	507.446	16	31.7154	8.5704
12	3	176	516	645.47	34	18.9844	6.8366
13	9	385	857	1397.99	52	26.8844	6.8891
14	10	228	391	368.291	17	21.6642	7.1254
15	2	207	584	363.35	17	21.3735	7.2011

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	262.096	5607	13819	0.5809	0.5584	0.0342429	0.0259787	1.5262
2	229.522	9151	23871	1	0.999	0.0585728	0.0550878	7.4067
3	120.811	6134	17911	1	1	0.0588523	0.0524817	4.9664
4	309.279	11331	27269	1	1	0.0420087	0.0470131	3.88
5	235.238	9387	21888	1	1	0.033344	0.0384686	2.9757
6	214.167	8637	19737	1	1	0.0500174	0.0454476	4.2023
7	322.836	9940	21386	0.8402	0.7724	0.0394366	0.0383896	3.4073
8	229.48	8595	22042	1	1	0.0437464	0.0406043	3.4861
9	156.853	6922	19429	1	1	0.0309159	0.0332493	5.7379
10	109.498	4245	11066	1	0.9865	0.0409894	0.042111	4.5663
11	161.817	5491	13376	1	1	0.0346021	0.0355861	3.0899
12	147.196	6914	19480	1	1	0.0254556	0.0264887	2.0381
13	254.693	9728	23664	1	1	0.0395765	0.0362154	3.5337
14	202.693	7711	16371	0.9376	0.8055	0.0295682	0.0238837	4.9336
15	47.4558	4187	11845	1	1	0.0494387	0.0493035	4.2144

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	85	150	219.51	395.664	2	6	6
2	528	1263	590.333	1486.51	12	17	18
3	319	940	407.806	1046.07	8	11	11
4	450	1148	539.304	1444.7	13	14	16
5	278	702	338.409	926.056	7	12	13
6	361	705	481.534	1004.52	9	11	15
7	379	769	440.563	924.896	10	12	12
8	358	644	417.664	999.11	10	12	10
9	191	482	252.037	750.382	5	7	9
10	161	377	194.51	519.99	6	5	6
11	190	422	220.238	534.647	5	7	8
12	127	316	199.995	597.085	5	7	8
13	342	639	432.363	959.838	9	13	15
14	189	249	254.723	437.233	6	5	7
15	145	428	227.782	658.902	3	5	10

682_z29: distributions, pre-validation, px